QPCODE: 25900285

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCHE202 - INORGANIC AND ORGANIC CHEMISTRY

Instruction to the Examiners:

- *This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.*
- *Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.*
- *Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.*

Part A
Multiple choice questions, answer all questions, each question carries 1 mark

1. In vitamin B_{12}, cobalt is coordinated in a:
   Ans: b). Corrin ring
2. A food product label shows that it contains BHA. What purpose does it serve in the food?
   c). Prevents oxidation of fats
3. Why is pyrrole considered an aromatic compound despite having a lonepair of electrons on nitrogen and being a five-membered ring?
   d). All of the above
4. Which molecule provides energy for the sodium-potassium pump?
   Ans: a) ATP
5. The process of fruit ripening is chiefly regulated by the gas
   b). Ethylene
6. How many Optically active isomers are there for Lactic acid?
   Answer: b). 2
7. What is the main principle used in a nuclear power reactor?
   a). Fission
8. Stereoisomers of a compound which are not mirror images of each other are known as
   c). Diastereomers
9. A compound which has the ability to sustain an induced ring current is called
   a) Diatropic **(beyond the scope of the syllabus – give full mark)**
10. Fast foods are often high in:
    c). Saturated fats, and salt

Part B
Short Answer Questions- Answer any 4 questions, each question carries 3 marks

11. How can you detect the presence tamarind or date seed powder in coffee powder?
Tamarind or date seed powder(adulterants) contain more starch than pure coffee powder. Mix a small amount of coffee powder with water, allow it to settle, and then add a few drops of iodine solution. If the mixture turns blue or blue-black, the presence of starch and thus the adulterants is confirmed.
(Any valid three points – 3 marks)

12. Summarise the role played by hemoglobin in oxygen transport

- Hemoglobin binds with oxygen from the lungs. (Concentration of oxygen is more in lungs)
- It transports oxygen through the blood
- In tissues, it releases oxygen to myoglobin. (Concentration of oxygen is more in lungs)

(Any valid three points – 3 marks)

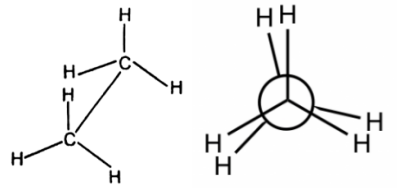
13. Explain Geometrical Isomerism of Aldoximes and Ketoximes

Beyond the scope of syllabus-give full marks to all

14. Explain Critical, Subcritical, and Supercritical Mass

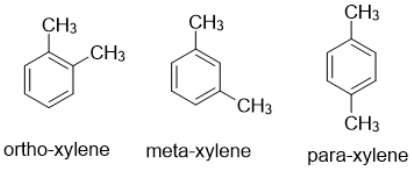
Subcritical Mass: A mass of fissile material that is too small to sustain a chain reaction. The rate of neutron leakage is greater than the rate of neutron production (K<1). **(Any valid point – 1 mark)**
Critical Mass: The minimum amount of fissile material needed for a sustained nuclear chain reaction. The rate of neutron leakage equals the rate of neutron production (K=1). **(Any valid point – 1 mark)**
Supercritical Mass: A mass of fissile material that is greater than the critical mass. The rate of neutron production is greater than the rate of neutron leakage (K>1), leading to an exponentially increasing reaction (used in weapons). **(Any valid point – 1 mark)**

15. Draw the Sawhorse and Newman Projections of Eclipsed Ethane



(two structures – 3 marks)

16. Draw different isomeric structures of Xylene



(three structures- 3marks)

Part C
Short Essay Questions- Answer any 2 questions, each question carries 7 marks

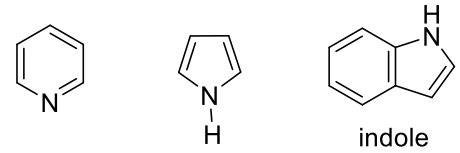
17. What are artificial sweeteners with examples and write the qualities of an ideal sweetener?
Artificial sweeteners (or sugar substitutes) are compounds used to sweeten foods and beverages without adding significant calories. They are much sweeter than sucrose (table sugar) Examples: Saccharin, cyclamate Aspartame, Alitame, Sucralose, Acesulfame K, Neotame. Qualities of an Ideal Sweetener are, must be non-toxic, non-caloric, non-carcinogenic, easily soluble in water, stable under processing and storage conditions, affordable and easily incorporated into food systems and should have a taste profile similar to sugar. Ideal sweetener has a multiple approach in which they reduce the cost and improve the product taste and stability.
(Definition-3 marks, one examples-1mark, 3 qualities-3 marks)

18. The cis isomer normally has a higher dipole moment than the corresponding trans isomer. Justify with suitable examples
Geometrical isomers (cis/trans) have different spatial arrangements of atoms, which affects the overall molecular polarity and thus the dipole moment (μ). The cis isomer generally has a higher dipole moment than the corresponding trans isomer because of the vector summation of individual bond dipoles.
E.g. Maleic acid (cis isomer) has a net dipole moment, while fumaric acid (trans isomer) has a zero-dipole moment. This is because in maleic acid, individual dipole moments of carboxylic acid group add up to their cis configuration, where as in fumaric acid the dipoles cancel each other out due to their trans configuration.
(Two valid points-4marks, justification using examples-3 marks)

19. **Differentiate the structural differences between hemoglobin and myoglobin.**
i) Hemoglobin – in blood – monomer, whereas myoglobin in tissues tetramer.
ii) Both contain iron porphyrin complexes in heme group
iii) Hemoglobin- 4 protein chains groups/ amino acids/polypeptide chains, Myoglobin – 1 protein chains groups/ amino acids/polypeptide chains
iv) Hemoglobin – can transport 4 O_2, Myoglobin – can store one O_2
v) Hemoglobin –cooperative binding, Myoglobin – No cooperative binding o O_2.
(Any 4 points with relevant explanation – 7 marks)

Part D
Essay type Questions- Answer any 1 question, each question carries 14 marks

20. Compare the basicity of pyrrole, pyridine, and indole. Explain with molecular orbital considerations and examples.



pyridine pyrrole indole

(Since the question is beyond the scope of the syllabus, give full marks if the structure of three compounds is drawn properly)

21. Elaborate on the use of radioisotopes in various fields.

Radioisotopes are atoms with unstable nuclei that decay spontaneously, releasing radiation such as alpha, beta, or gamma rays

1. Medicine (Nuclear Medicine):

Diagnosis (Tracers): Radioisotopes like Technetium-99m (Tc-99m) are used for imaging (e.g., SPECT scans, PET scans) to monitor organ function and blood flow. Uses *Fluorine-18* (^{18}F) in glucose molecules to detect cancer and monitor brain activity.

Therapy: High-energy radiation from isotopes like Iodine-131 (I-131) is used for thyroid cancer treatment

Cobalt-60 and *Cesium-137* emit gamma rays used in radiotherapy to destroy malignant cells. *Cesium-137* helps prevent graft-versus-host disease during blood transfusion

2. Agriculture and Food Safety

Pest Control: Irradiation of insects to control pests

Food Preservation: Gamma irradiation (e.g., from Cobalt-60) is used to kill bacteria and prolong shelf life of food

Mutation breeding: *Cobalt-60* and *Gamma rays* induce mutations to develop new high-yield and disease-resistant crop varieties

Fertilizer studies: *Phosphorus-32* and *Nitrogen-15* help study nutrient uptake by plants, optimizing fertilizer use

3. Industry

Non-Destructive Testing (NDT): Ir-192 or Co-60 are used in industrial radiography to detect flaws (cracks or voids) in welds and metals

Thickness and density measurement: *Strontium-90* and *Cesium-137* are used to control thickness of materials like paper, metal, and plastic sheets

Leak detection: *Tritium* (^{3}H) and *Sodium-24* are used to trace leaks in pipelines

4. Research

Tracers in Chemistry/Biology: Isotopes like Carbon-14 or Hydrogen-3 are used to trace reaction mechanisms or metabolic pathways, and biological processes

Radiolabelling helps in drug development and environmental studies (tracking pollutants)

5. **Archaeology and Geology**

Carbon dating: *Carbon-14* determines the age of archaeological artifacts up to ~50,000 years old.

Uranium–lead dating: *Uranium-238* and *Lead-206* used to date rocks and minerals billions of years old.

Environmental tracing: *Tritium* and *C-14* are used to study groundwater movement and sedimentation rates

6. Power Generation

Nuclear reactors use *Uranium-235* and *Plutonium-239* as fuel to produce energy through controlled fission

Radioisotope thermoelectric generators (RTGs) use isotopes like *Plutonium-238* to provide long-term power for space missions (e.g., Voyager, Curiosity rover)

(Any 4 uses with explanation, each use gives 3.5 marks)
